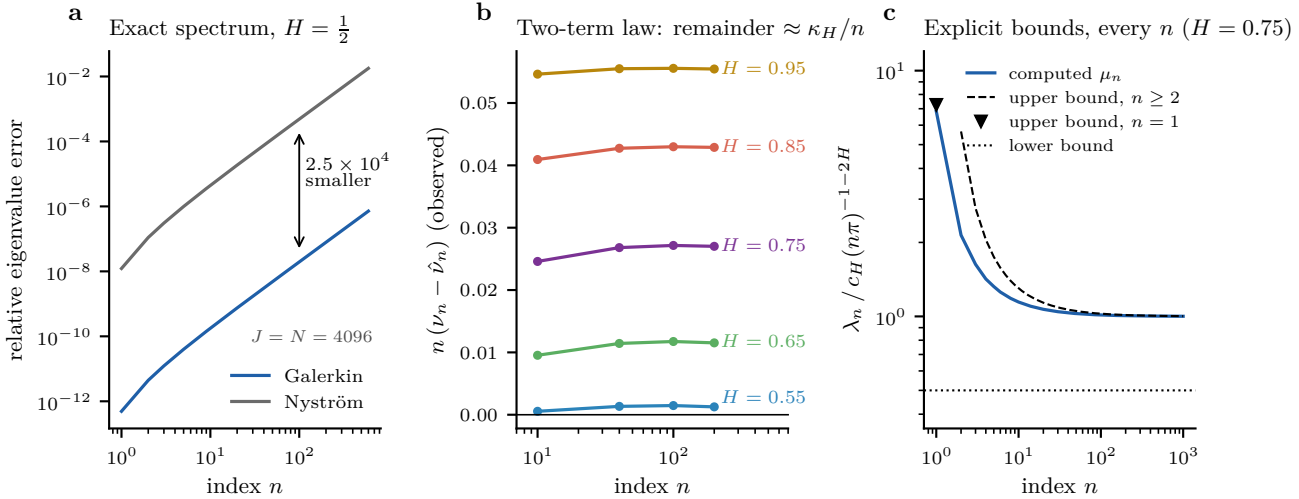

a
Exact spectrum, $H = \frac{1}{2}$
relative eigenvalue error
10^{-2}
10^{-4}
10^{-6}
10^{-8}
10^{-10}
10^{-12}
2.5×10^4 smaller
$J = N = 4096$
Galerkin
Nyström
10^0
10^1
10^2
index n
b
Two-term law: remainder $\approx \kappa_H / n$
$n\,(\nu_n - \hat{\nu}_n)$ (observed)
0.05
0.04
0.03
0.02
0.01
0.00
$H = 0.95$
$H = 0.85$
$H = 0.75$
$H = 0.65$
$H = 0.55$
10^1
10^2
index n
c
Explicit bounds, every n ($H = 0.75$)
$\lambda_n \,/\, c_H\,(n\pi)^{-1-2H}$
10^1
10^0
computed μ_n
upper bound, $n \geq 2$
upper bound, $n = 1$
lower bound
10^0
10^1
10^2
10^3
index n